

State-Space Essentials

Item	Continuous time	Discrete time
Model	$\dot{x} = Ax + Bu, y = Cx + Du$	$x(k+1) = Gx(k) + Hu(k)$ $y(k) = Cx(k) + Du(k)$
Dimensions	$A: n \times n, B: n \times m, C: p \times n, D: p \times m$	$G: n \times n, H: n \times m, C: p \times n, D: p \times m$
TF	$G(s) = C(sI - A)^{-1}B + D$	$G(z) = C(zI - G)^{-1}H + D$
Poles	$\det(sI - A) = 0$	$\det(zI - G) = 0$
Stability	all $\text{Re}(\lambda(A)) < 0$	all $ \lambda(G) < 1$

State = minimum information needed to predict future behaviour for known input. Internal stability checks all state modes; BIBO stability checks input-output poles only. Pole-zero cancellation can hide internally unstable modes. Block rules: series G_1G_2 , parallel $G_1 + G_2$, negative feedback $G_1/(1 + G_1G_2)$, positive feedback $G_1/(1 - G_1G_2)$.

Response formulas:

$$x(t) = e^{At}x_0 + \int_0^t e^{A(t-\tau)} Bu(\tau) d\tau.$$

$$\text{If } A = T\Lambda T^{-1}, e^{At} = Te^{\Lambda t}T^{-1}.$$

Lyapunov CT test: solve $A^TP + PA = -Q$; if $Q > 0$ and $P > 0$, asymptotically stable.

Eigenvalue response memory: real negative = exponential decay; real positive = exponential growth; complex pair = oscillation; negative real complex = decaying oscillation; positive real complex = growing oscillation; zero or pure imaginary = marginal/integrator behaviour. DT poles near origin = fast decay; poles near unit circle = slow decay/oscillation; negative DT pole = alternating sign.

2x2 inverse shortcut: if $M = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, then $M^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$. Useful for $C(sI - A)^{-1}B$ and $C(zI - G)^{-1}H$.

Mini example - TF/stability: $A = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}, B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, C = [1 \ 0]$. $\det(sI - A) = s^2 + 3s + 2 = (s + 1)(s + 2)$, so poles $-1, -2$ and stable. $G(s) = 1/(s^2 + 3s + 2)$.

Linearisation Workflow

For $\dot{x} = f(x, u), y = g(x, u)$:

- 1. Find equilibrium: $0 = f(\bar{x}, \bar{u}), \bar{y} = g(\bar{x}, \bar{u})$.
- 2. Define increments: $\delta x = x - \bar{x}, \delta u = u - \bar{u}, \delta y = y - \bar{y}$.
- 3. Compute Jacobians at the chosen equilibrium:

$$A = \left. \frac{\partial f_i}{\partial x_i} \right|_{\bar{x}, \bar{u}}, \quad B = \left. \frac{\partial f_i}{\partial u_i} \right|_{\bar{x}, \bar{u}}, \quad C = \left. \frac{\partial g}{\partial x} \right|_{\bar{x}, \bar{u}}, \quad D = \left. \frac{\partial g}{\partial u} \right|_{\bar{x}, \bar{u}}$$

- 4. Write $\delta \dot{x} = A\delta x + B\delta u, \delta y = C\delta x + D\delta u$.

Traps: do not linearise at the origin unless it is the specified equilibrium; linearisation is local; different equilibria can produce different stability. Linearity test: additivity and homogeneity.

Mini examples:

$$\text{If } \dot{x} = x^2 + u, y = x, \bar{x} = 2,$$

$$\text{then } \bar{u} = -4, A = 2\bar{x} = 4, B = 1, C = 1, D = 0.$$

$$\text{If } \dot{x}_2 = -(g/l)\sin x_1 + u,$$

$$\text{then } \partial \dot{x}_2 / \partial x_1 = -(g/l)\cos \bar{x}_1;$$

$$\text{at } 0 \text{ gives } -g/l, \text{ at } \pi \text{ gives } +g/l.$$

Mechanical modelling pattern: choose states as position and velocity.

$$\text{For } m\ddot{y} + b\dot{y} + ky = u,$$

let $x_1 = y, x_2 = \dot{y}$, so $\dot{x}_1 = x_2, \dot{x}_2 = -(k/m)x_1 - (b/m)x_2 + (1/m)u, y = [1 \ 0]x$. For electrical RC: state often capacitor voltage; use KCL and component laws to isolate $\dot{x}$.

Discrete Modelling, Sampling, Z-Plane

Exact ZOH discretisation for $\dot{x} = Ax + Bu$:

$$G = e^{AT}, \quad H = \int_0^T e^{A\lambda} B d\lambda, \quad y(k) = Cx(k) + Du(k)$$

If A invertible: $H = A^{-1}(G - I)B$.

If A singular: use integral, augmented matrix exponential, Laplace method, or series $H = (IT + AT^2/2 + A^2T^3/6 + \dots)B$.

Euler approximation: $G \approx I + AT, H \approx BT$ only for small T .

Mapping: $z = e^{sT}, s = (1/T)\ln z$. CT LHP maps inside unit circle;

imaginary axis maps to unit circle;

RHP maps outside. For $s = \sigma \pm j\omega, z = e^{\sigma T}e^{\pm j\omega T}$, so radius $e^{\sigma T}$ and angle ωT .

Sampling: $T = 1/f_s$. Nyquist: $f_s > 2f_{max}$. Practical control: usually $f_s \geq 10f_{bw}$, often 20 to 40 times bandwidth. Too slow = aliasing, delay, poor tracking/rejection. Too fast = noise sensitivity, computation/hardware cost.

Mini examples: $\dot{x} = -2x + u, T = 0.1$:

$$G = e^{-0.2} = 0.8187, H = (-1/2)(0.8187 - 1) = 0.0906.$$

$$\text{Pole } s = -4, T = 0.1: z = e^{-0.4} = 0.6703.$$

$$\text{Poles } s = -2 \pm j5, T = 0.1: \text{radius } 0.8187, \text{angle } \pm 0.5 \text{ rad.}$$

Difference equation to pulse TF: apply $Z\{y[k - i]\} = z^{-i}Y(z)$ assuming zero initial conditions. Example: $y[k] - 0.5y[k - 1] = 2u[k - 1]$ gives $Y(z)(1 - 0.5z^{-1}) = 2z^{-1}U(z)$, so $G(z) = Y/U = 2z^{-1}/(1 - 0.5z^{-1}) = 2/(z - 0.5)$.

Common z-transform facts: delay by one sample multiplies by z^{-1} ; poles inside unit circle decay; pole at $z = 1$ is an integrator/step-memory mode; pole at $z = -1$ gives alternating marginal oscillation.

Controllability, Observability, Final Value

Test	Matrix	Condition	Meaning
DT controllability	$C = [H \ GH \ \dots \ G^{n-1}H]$	$\text{rank}(C) = n$	input can move all modes
DT observability	$O = \begin{bmatrix} C \\ CG \\ \vdots \\ CG^{n-1} \end{bmatrix}$	$\text{rank}(O) = n$	outputs reveal all modes
CT versions	replace (G, H, C) with (A, B, C)	same rank tests	continuous models

If square, non-zero determinant implies full rank. Use controllability for state-feedback design; use observability for observer design. Discrete final value theorem: if stable, $\lim_{k \rightarrow \infty} y[k] = \lim_{z \rightarrow 1} (1 - z^{-1})Y(z)$.

Mini example - controllability: $G = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, H = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, GH = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, so $C = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$, $\det = -1 \neq 0$: controllable.

Observability mini example: for $G = \begin{bmatrix} 1 & 0.1 \\ 0 & 0.9 \end{bmatrix}$ and $C = [1 \ 0]$, $CG = [1 \ 0.1]$, so $O = \begin{bmatrix} 1 & 0 \\ 1 & 0.1 \end{bmatrix}$ has determinant $0.1 \neq 0$: observable. If $C = [0 \ 1]$, then $CG = [0 \ 0.9]$ and rank is one: not observable.

Exam conclusion wording: after a rank/eigenvalue test, write the condition and conclusion in words. Example: $\text{rank}(C) = 2 = n$, therefore all modes are controllable and arbitrary DT pole placement is possible.

State Feedback, Tracking, Integral Action

State feedback: $u(k) = -Kx(k)$, closed-loop matrix $G_{cl} = G - HK$. Closed-loop poles are $\text{eig}(G - HK)$. Pole placement workflow: check controllability; choose DT poles inside unit circle; convert CT poles with $z = e^{sT}$ if needed; match $\det(zI - (G - HK))$ to desired polynomial; verify response and control effort.

Pole placement template: for $G = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$, $H = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, $K = [k_1 \ k_2]$, $G - HK = \begin{bmatrix} 1 & 1 \\ -k_1 & 1 - k_2 \end{bmatrix}$.

Match $\det(zI - (G - HK))$ to $(z - z_1)(z - z_2)$.

Pole choice guide: poles closer to origin = faster response but larger control effort; poles closer to unit circle = slower response but lower effort; complex poles = oscillatory response; repeated poles can create sensitivity. Always check actuator saturation after placing fast poles.

Tracking vs regulation: regulation drives output/state to zero; tracking drives output to r . Constant-reference tracking can use $u = K_r r - Kx$. Integral action is more robust for unknown constant disturbances. Internal model principle: to track/reject a signal asymptotically, controller must include that signal's dynamics: step = one integrator; ramp = two integrators; sinusoid = oscillator.

Reference feedforward: for SISO $D = 0$, steady-state constant tracking with $u = K_r r - Kx$ can use $K_r = 1/[C(I - G + HK)^{-1}H]$ when the closed loop is stable and the inverse exists. If model mismatch/disturbance exists, prefer integral action.

Integral augmentation examples:

$$q(k+1) = q(k) + r(k) - y(k) \quad \text{or} \quad q(k+1) = q(k) + y(k)$$

For output-integrator convention:

$$z = \begin{bmatrix} x \\ q \end{bmatrix}, \quad G_z = \begin{bmatrix} G & 0 \\ C & I \end{bmatrix}, \quad H_z = \begin{bmatrix} H \\ 0 \end{bmatrix}, \quad u = -K_z z = -K_x x - K_I q$$

Check augmented controllability before placing augmented poles. Integral action increases system order and can increase overshoot/control effort.

Sinusoidal internal model: a sinusoid with discrete frequency ω can be generated by a second-order oscillator with characteristic $z^2 - 2\cos(\omega)z + 1$. Add oscillator states if sinusoidal tracking/rejection is required.

Disturbance feedforward idea: if a measurable disturbance d enters through H_d , use a feedforward term to cancel its steady-state effect. If the disturbance is unmeasured or model is uncertain, integral action is usually more robust.

DLQR And MPC

DLQR minimises infinite-horizon quadratic cost:

$$J = \sum_{k=0}^{\infty} [x(k)^T Q x(k) + u(k)^T R u(k)], \quad Q \succcurlyeq 0, R \succ 0$$

DARE and gain:

$$P = G^T P G - (G^T P H)(R + H^T P H)^{-1}(H^T P G) + Q, \quad K = (R + H^T P H)^{-1} H^T P G$$

Stability theorem: if (G, H) controllable and (G, V) observable where $Q = V^T V$, then the stabilising solution gives stable $G - HK$. Note: (G, V) is cost observability, not sensor observability (G, C) .

Tuning sheet: larger q_{ii} penalises state x_i more; larger r_{jj} penalises input u_j more. Larger Q/R = faster, more aggressive, more input. Larger R/Q = slower, less input. Scaling both Q and R by the same factor gives same K . Bryson rule: $q_{ii} \approx 1/x_{i,max}^2$, $r_{jj} \approx 1/u_{j,max}^2$. If too slow: increase relevant Q . If saturated/noisy input: increase R . LQR does not explicitly handle constraints.

MPC: finite-horizon optimisation solved online each sample; applies first input only, then re-solves. Handles input/state/rate constraints and saturation directly. More computational cost than LQR.

MPC cost template: minimise $\sum_{j=0}^{N-1} (x_{k+j}^T Q x_{k+j} + u_{k+j}^T R u_{k+j}) + x_{k+N}^T P_f x_{k+N}$ subject to $x_{k+j+1} = Gx_{k+j} + Hu_{k+j}$ and constraints. Larger horizon N improves planning but increases computation.

Observers, Output Feedback, Kalman Filters

Open-loop predictor $\hat{x}(k+1) = G\hat{x}(k) + Hu(k)$ has error $e(k+1) = Ge(k)$, so unstable or slow plants give bad estimates. Luenberger observer:

$$\hat{x}(k+1) = G\hat{x}(k) + Hu(k) + L[y(k) - C\hat{x}(k)]$$

Error $e = x - \hat{x}$ follows $e(k+1) = (G - LC)e(k)$. Observer design workflow: check observability of (G, C) ; choose observer poles inside unit circle; make them faster than controller poles but not too fast; compute

Output feedback: $u = -K\hat{x}$. Separation principle:

$$\begin{bmatrix} x(k+1) \\ e(k+1) \end{bmatrix} = \begin{bmatrix} G - HK & HK \\ 0 & G - LC \end{bmatrix} \begin{bmatrix} x(k) \\ e(k) \end{bmatrix}$$

Overall poles = controller poles $\text{eig}(G - HK)$ plus observer poles $\text{eig}(G - LC)$. Design K and L independently for ideal models. Fast observer poles give fast convergence but amplify measurement noise.

Kalman model: $x(k+1) = Gx(k) + Hu(k) + w(k)$, $y(k) = Cx(k) + v(k)$. $Q = E[ww^T]$ process/model-noise covariance; $R = E[vv^T]$ measurement-noise covariance. Larger Q/R = trust measurements more; estimates respond faster but noisier. Smaller Q/R = trust model more; smoother but may drift or lag. Residual $r(k) = y(k) - C\hat{x}^-(k)$ should be zero-mean, white, and expected variance. Non-zero mean = bias; correlated residuals = model/tuning mismatch. EKF linearises nonlinear f, h about current estimate each step; useful but approximate and more computationally expensive.

Kalman predict-correct skeleton: predict $\hat{x}^- = G\hat{x} + Hu$, $P^- = GPG^T + Q$; update $K_k = P^- C^T (CP^- C^T + R)^{-1}$, $\hat{x} = \hat{x}^- + K_k(y - C\hat{x}^-)$, $P = (I - K_k C)P^-$. Luenberger chooses L by poles; Kalman computes gain from uncertainty.

Observer mini example: if controller poles are 0.7, 0.8, observer poles 0.2, 0.3 are faster. If sensor noise dominates, move observer poles less aggressively toward zero.

Controller Discretisation By Emulation

Direct design: discretise plant, design controller in z domain. Emulation: design $G_c(s)$ first, convert to $G_d(z)$.

Method	Rule	Main use / warning
Tustin/bilinear	$s \leftarrow \frac{2}{T} \frac{z-1}{z+1}$	preserves stability; frequency warping
ZOH equivalent	$G_d(z) = (1 - z^{-1})Z\{G_c(s)/s\}$	exact at sample instants for staircase input; phase lag
Matched pole-zero	map each pole/zero with $z = e^{sT}$, match $G_d(1) = G_c(0)$	preserves pole/zero locations; add zeros at $z = -1$ if needed

Emulation quick example: continuous pole $s = -a$ maps by MPZ to $z = e^{-aT}$. For Tustin, substitute $s = (2/T)(z-1)/(z+1)$ directly into $G_c(s)$, then simplify and normalise powers of z or z^{-1} .

High-Yield Traps And Memory List

CT stability uses $\text{Re}(s) < 0$; DT stability uses $|z| < 1$. Convert CT poles to DT poles before discrete pole placement. Do not use $H = A^{-1}(G - I)B$ if A is singular. Exact ZOH usually keeps C, D unchanged. Controllability is for moving states with inputs; observability is for reconstructing states from outputs. BIBO stable does not guarantee internally stable if unstable modes are hidden. Integral action changes system order and requires augmented controllability. LQR Q = state penalty; Kalman Q = process noise. DLQR theorem uses (G, V) with $Q = V^T V$, not (G, C) . Standard LQR does not handle saturation; MPC does. Very fast observer poles amplify noise.

Minimum formulas to memorise: $C(sI - A)^{-1}B + D$, $G = e^{AT}$, $H = \int_0^T e^{A\lambda} B d\lambda$, $z = e^{sT}$, $C = [H \ GH \ \dots \ G^{n-1}H]$, $\mathcal{O} = [C; CG; \dots; CG^{n-1}]$, $G - HK$, $G - LC$, $K = (R + H^T P H)^{-1} H^T P G$.

Fast exam templates: SS to TF: form $sI - A$, invert, multiply $C(\cdot)B + D$. Linearise: equilibrium, perturbations, Jacobians, incremental model. Discretise: compute G , compute H , map poles. Pole placement: controllability, target polynomial, coefficient matching. Observer: observability, target observer poles, $L = \text{place}(G', C', p)'$. LQR/Kalman concept question: define what Q, R mean before describing tuning.